Trainings

General Information

Cummins Inc. certifies its engines using the prescribed EPA and European Certification Fuels. Cummins Inc. does **not** certify engines on any other fuel. It is the user's responsibility to use the correct fuel as recommended by the manufacturer and allowed by EPA or other local regulatory agencies. In the United States, EPA allows **only** registered fuels and fuel additives to be entered into commerce. EPA has additional alternative fuel information at:

- https://www.epa.gov/renewable-fuel-standard-program/alternative-fuels (https://www.epa.gov/renewable-fuel-standard-program/alternative-fuels)

Biodiesel Terminology:

- Biofuels - Fuels produced from renewable resources.
- Biodiesel - A fuel comprised of methyl or ethyl ester-based oxygenates of long chain fatty acids derived from the transesterification of vegetable oils, animal fats, and cooking oils. These fuels are commonly known as Fatty Acid Methyl Esters (FAME) or Fatty Acid Ethyl Esters (FAEE). Biodiesel properties are similar to those of diesel fuel, as opposed to gasoline or gaseous fuels, and thus are capable of being used in compression ignition engines.
- B100 - A fuel containing 100 percent biodiesel.
- Petrodiesel - Diesel fuel produced purely from petroleum. Petrodiesel can also be referred to as distillate diesel.
- Biodiesel Blend - A fuel comprised of a mixture of petrodiesel and pure biodiesel. A biodiesel blend is typically designated by the percentage of biodiesel in the blend. For example: B5 is a fuel containing 95 percent petrodiesel and 5 percent B100.
- Rapeseed Methyl Ester (RME) diesel - Biodiesel derived from rapeseed oil. RME diesel is the most common biodiesel used in Europe.
- Soy Methyl Ester (SME or SOME) diesel - Biodiesel derived from soybean oil. SME diesel is the most common biodiesel used in the United States.
- BQ-9000 - The National Biodiesel Accreditation Program, which is called BQ-9000, is a cooperative and voluntary program for the accreditation of producers and marketers of biodiesel fuel. The program is a unique combination of the ASTM standard for biodiesel, ASTM D6751, and a quality systems program that includes storage, sampling, testing, blending, shipping, distribution, and fuel management practices.

With increased interest in reducing the use of petroleum distillate fuels, many governments and regulating bodies encourage the use of biofuels, such as biodiesel.

Cummins Inc. test data on the operating effects of biodiesel fuels indicates that typically smoke, power, and fuel economy are all reduced.

There are multiple specifications for biodiesel issued by The European Committee for Standardization (CEN) and the American Society for Testing & Materials (ASTM). These specifications are utilized worldwide and govern the quality of biodiesel and usage of the fuel. See Table 4.

Table 4, Common Biodiesel Specifications

Standard Organization	**Specification Number**	**Biodiesel Range Percentage**	**Usage**
CEN	EN14214	100	Finished Fuel or Blend Component
CEN	EN16709	30 and 20	Finished Fuel
CEN	EN16734	10	Finished Fuel
ASTM	D6751	100	Blend Component
ASTM	D7467	6-20	Finished Fuel

⚠ CAUTION ⚠

Cummins Inc. only approves biodiesel blends up to a maximum of B20 for use in certain engines. See Table 6 of the section for further guidance.

⚠ CAUTION ⚠

To successfully use biodiesel, it is imperative that the fuel be of high quality and meet or exceed the specifications outlined in this bulletin or engine damage will occur.

It is the responsibility of the user to verify/obtain the proper local, regional, or national exemptions required for the use of biodiesel in any emissions regulated Cummins® engine.

Warranty and the Use of Biodiesel Fuel in Cummins® Engines

Cummins Inc. Engine Warranty covers failures that are a result of defects in material or factory workmanship. Engine damage, service issues, and/or performance issues determined by Cummins Inc. to be caused by the use of biodiesel fuel **not** meeting the specifications outlined in this Service Bulletin are **not** considered to be defects in material or workmanship and are **not** covered under Cummins Inc. engine warranty.

Requirements for Using Biodiesel Fuel in Cummins® Engines

Cummins Inc. requires that all biodiesel fuel blends be comprised of petrodiesel meeting ASTM D975, and biodiesel meeting either ASTM D6751 or EN14214. Diesel fuel and biodiesel blends up to B7 **must** meet the specifications found in Table 1: Cummins Inc. Required Diesel Fuel Specifications. For biodiesel blends above B6 and up to B20, Cummins Inc. requires that the fuel meet the specifications outlined in ASTM D7467. These specifications are summarized in Table 7: Summary of ASTM D7467 Requirements for B6 to B20 Biodiesel Blends. Reference the official ASTM D7467 standard for more detailed requirements.

All Cummins® engines are compatible with biofuel blends up to B7. Engines that are compatible with biofuel blends up to B20 are listed in Table 6 of Cummins® Fuels Bulletin. Engines that are built after 1st January 2008 are compatible with biofuel blends up to B20, except the engines that are listed in Table 5. Engines listed in Table 6 that are manufactured prior to January 2008 and have been rebuilt after January 2008 are considered B20 compatible.

For assistance in identifying the associated service model name:

- See Technical Service Bulletin, Cummins® Service Engine Model Identification, TSB130080. Review the associated Cummins® Product Technology procedures.
    - https://quickserve.cummins.com/qs3/pubsys2/xml/en/tsb/2013/tsb130080.html (https://quickserve.cummins.com/qs3/pubsys2/xml/en/tsb/2013/tsb130080.html)
- Use QuickServe™ Online and enter the engine serial number for the engine being serviced. QuickServe™ Online will display the service model name associated with the engine serial number entered.

Table 5, Acceptable Use Up to B7 in accordance with ASTM D7467		
Product Series	**Displacement (Liters)**	**Products**
K, T, QSK, QST	19, 23, 30, 38, 50, 60, 78, 95	Engines built before January 2008 or engines equipped with aftertreatment
QSK	50	Engines equipped with dual fuel conversion

Biodiesel fuel can be blended with an acceptable diesel fuel up to a 20 percent volume concentration (B20) for the following Cummins® engines:

Table 6, Acceptable Use Up to B20			
Product Series	**Displacement (L)**	**Products**	**Notes**
F	2.8, 3.8, 4.5	All engines built after January 2008	
V	5	ISV5.0 CM3230 V104	
B/D	3.3, 3.9, 4.5, 5.9, 6.7	All engines built after January 2007.	a,b
		Marine engines produced after 01 January 2007: B-Series & QSB.	

Table 6, Acceptable Use Up to B20

Product Series	Displacement (L)	Products	Notes
C/L	8.3, 8.5, 9, 10	All engines built after January 2007.	a,b
		Marine engines produced after 01 January 2007: C-Series, QSC, and QSL.	
		8.5L and 10L engines produced by Tata-Cummins Private Limited.	
M	11	All engines built after January 2007.	
		Marine engines produced after 01 January 2007: QSM11.	
G	11, 12	All engines built after January 2007.	
N	14	All engines built after January 2007.	
X	11.9, 15	ISX CM570 built after January 2002.	
		All engines built after January 2007.	
K, QSK, QST	19, 23, 30, 38, 45, 50, 60, 78, 95	All engines built after January 2008 without aftertreatment.	a,b,c
QSK	50	Dual fuel **not** approved on B20.	
QSK	95	Tier 4 Rail applications **only**	
Z	13	All engines produced by Dongfeng-Cummins Engine Company after January 2007	

Table 6, Acceptable Use Up to B20			
Product Series	**Displacement (L)**	**Products**	**Notes**
Z	14	All engines produced by Dongfeng-Cummins Engine Company after January 2020	
M	15	All engines produced by Dongfeng-Cummins Engine Company after January 2021	
M	8.5, 10	All engines produced by Dongfeng-Cummins Engine Company.	
a: If equipped with Eliminator™ oil change extender system, Cummins Inc. requires oil sampling. See details below. b: For marine applications, Cummins Inc. requires additional fuel water separation capability due to potential ballast water mixing with fuel. See additional fuel water separation details below. c: For these HHP engines built after January 2008 WITHOUT an Aftertreatment system are approved for B20 fuel. Then for the same engines built WITH an Aftertreatment system are approved for B7 fuel.			

To avoid confusion with acceptable use of biodiesel between various engine names, enter the ESN in question into QSOL to determine the specific service engine model name and build date.

Customers choosing to run biodiesel blends above B7 and up to B20 **must** adhere to the following requirements from Cummins Inc.

Note : For North American markets, Cummins Inc. requires that the biodiesel fuel blend be purchased from a BQ-9000 Certified Marketer. The B100 biodiesel fuel used in the blend must be sourced from a BQ-9000 Accredited Producer. Certified Marketers and Producers can be found at the following website: http://www.bq-9000.org. For areas outside of North America, consult a Cummins Inc. representative for applicable fuel quality standards.

Oil Sampling:

- Fuel dilution of lubricating oil has been observed with the operation of biodiesel under certain operating conditions. Fuel dilution monitoring can be accomplished by performing oil sampling. Fuel levels in lubricating oil **must not** exceed 5 percent. Additional information on oil contamination and oil sampling can be found in Cummins® Engine Oil and Oil Analysis Recommendations, Bulletin 3810340 (/qs3/pubsys2/xml/en/bulletin/3810340.html).
- For ISB CM2150 and ISC/ISL CM2150 products, end users are **required** to use oil sampling during the first 6 months of operation with biodiesel to monitor engine oil condition and fuel

dilution of lubricating oil in order to determine if the oil change interval needs to be modified. Consult a Cummins® Authorized Repair Location for guidance in oil sampling.

- For High Horsepower engines equipped with the Eliminator™ oil filter option, oil sampling will be needed to determine the appropriate oil change interval. Oil samples should be taken every 250 hours of operation and analyzed according to the Cummins® Engine Oil Recommendations Bulletin. This process should be repeated for at least three oil change intervals to ensure consistent oil behavior.

Fuel-Water Separation:

- Biodiesel has a natural affinity to water, and water accelerates microbial growth. Storage tanks **must** be equipped with a fuel water separator to make sure that water is stripped out before entering the vehicle tank. Make sure the vehicle and storage tanks are kept full to reduce the potential for condensation accumulating in the fuel tank.
- Due to the solvent nature of biodiesel, and the potential for "cleaning" of the vehicle fuel tank and lines, new fuel filters **must** be installed when switching to biodiesel on used engines. Fuel filters will need to be replaced at half the standard interval for the next two fuel filter changes.
- Cummins Inc. requires the use of a StrataPore™ fuel filter media, and strongly recommends using Fleetguard® filters equipped with StrataPore™ media. This filter media removes water more efficiently than standard cellulosic filter media, which will **not** provide adequate fuel water separation capabilities. However, even StrataPore™ fuel filter media is **not** as effective in removing water from biodiesel as it is in removing water from petrodiesel. Therefore, preventing water from entering the fuel supply (vehicle or storage) remains very important.
- If StrataPore™ filter media is **not** available, a substitute synthetic filter media may be used which **must** provide 95 percent emulsified fuel water separation efficiency per SAE J1488. This test method **must** be run using B20 biodiesel, having an interfacial surface tension of 22 dyne/cm + or - 2 dyne/cm. The filter **must** meet this specification when run at the rated flow of the engine platform's fuel system. Fuel filter gaskets **must** also be compatible with B20 biodiesel blends, with performance equal to or greater than what is outlined in the Atmus™ Filtration Technologies Inc. Engineering Standard FES1544 - Seals, Static, Rubber (Supplier Requirements, Fuel Applications).
- For all High Horsepower Marine applications, a centrifuge filtration system is required to safeguard against water contamination. A centrifuge is recommended for all Commercial Marine applications. Water intrusion is common in vessel fuel storage tanks. Vessels may have multiple fuel storage tanks depending on type of service. Typically, one or two of the storage tanks are dedicated for engine supply. Fuel is transferred from the storage tanks to one or more day tanks. All engines are fueled from the day tanks. The centrifuge should be installed between the storage tanks and day tanks and should circulate fuel continuously. The target deliverable fuel quality of a centrifuge should meet or exceed the specifications outlined in ISO 4406: 18/16/13 (reference the "Fuel Cleanliness" section for more detail) and should have a maximum of 200 ppm dissolved or emulsified water, with no free water permitted.
- Cummins Inc. does **not** endorse specific suppliers of centrifuges, however Alfa Laval and Westfalla are two known manufacturers of Marine filtration devices. The Alfa Laval models are MAB 103, 104, and 206. Models MMB 304 and 305 are self cleaning units. The smallest version cleans 1135 l/hr (300 gal/hr), while larger versions can have flow rates over 10,000

l/hr (2,600 gal/hr). Most of these centrifuges clean fuel to the 3 to 5 micron range, but it takes up to 13 tank cycles to achieve this cleanliness. Some Westfalla models are the OTC 2/3-02-137 and OTC 2/3-03-107. They range from 500 l/hr (130 gal/hr) to 800 l/hr (215 gal/hr). These also clean fuel to the 3 to 5 micron range, but still require cycling the fuel in the tank up to 13 times. Refer to the manufacturer's product specifications and installation instructions.
- Fleetguard® Fuel Pro®, Diesel Pro®, Industrial Pro™, and Sea Pro®, products offered by Fleetguard® /Davco® Technology LLC, can be used to provide remote mounted additional fuel filtration efficiency, with integrated fuel pre-heaters. Consult a Cummins® Authorized Repair Location for guidance in fuel filter selection and installation.

Biodiesel Fuel Storage:

- Use biodiesel fuel within six months of its manufacture. Biodiesel has poor oxidation stability, which can result in long term storage problems. For this reason, Cummins Inc. does **not** recommend using biodiesel for low use applications, such as standby power, recreational marine, or seasonal applications. Consult your fuel supplier for oxidation stability additives.
- The poor oxidation stability qualities of biodiesel can accelerate fuel oxidation in the fuel system, especially at increased ambient temperatures.

⚠ CAUTION ⚠

Avoid storing equipment with biodiesel blends in the fuel system for more than three months, or fuel system damage can occur.

- If biodiesel is used for seasonal applications, the fuel system **must** be purged before storage by running the engine on pure diesel fuel for a minimum of 30 minutes.
- Care **must** also be taken when storing biodiesel in bulk storage tanks. All storage and handling systems **must** be properly cleaned and maintained. Steps must be taken to minimize moisture and microbial growth in storage tanks. Consult your fuel supplier for assistance in storing and handling biodiesel.

Energy Content:

- Although **not** approved for normal operation, Cummins Inc. has performed internal testing on B100 biodiesel and found that the fuel provides approximately 7 percent to 10 percent less energy per gallon of fuel when compared to conventional diesel fuels. Operation with B20 biodiesel blends can potentially result in a slight decrease in fuel economy and/or power, depending on the application. To avoid engine problems when the engine is converted back to 100 percent petrodiesel, do **not** change the engine rating to compensate for the potential power loss when operated with biodiesel fuels.

Materials Compatibility:

- The engines listed in this bulletin are compatible with biodiesel blends up to B20. However, the following **must** be taken into account:
- Natural rubber, butyl rubber, and some types of nitrile rubber (depending on chemical composition, construction, and application) may be particularly susceptible to degradation.

Also, copper, bronze, brass, tin, lead, and zinc can cause deposit formations. The use of these materials and coatings should be avoided for vehicle fuel tanks and fuel lines.

⚠ CAUTION ⚠

Contact your vehicle manufacturer to determine if any of the OEM supplied components are at risk with biodiesel in order to prevent engine damage.

Low Temperature Performance:

- Biodiesel fuel properties change at low ambient temperatures, which can pose problems for both storage and operation. Precautions can be necessary at low ambient temperatures, such as storing the fuel in a heated building or a heated storage tank, or using cold temperature additives.
- The fuel system can require heated fuel lines, filters, and tanks. Filters can plug and fuel in the tank can solidify at low ambient temperatures if precautions are **not** taken. A fuel heater is recommended for ambient temperatures below -5°C [23°F]. Consult your fuel and additive supplier for assistance in attaining proper cloud point fuel.

Microbial Growth:

- Biodiesel fuel is an excellent medium for microbial growth. Microbes cause fuel system corrosion and premature filter plugging. The effectiveness of all commercially available conventional anti-microbial additives, when used in biodiesel, is **not** known. Consult your fuel and additive supplier for assistance.

It is strongly recommended that customers running biodiesel blends of B7 or below follow the above precautions as well.

Table 7, Summary of ASTM D7467 Requirements for B6 to B20 Biodiesel Blends.

Item	Performance Characteristics	Requirements		Test Procedure
- - -	- - -	D1 Blends	D2 Blends	- - -
1	Flash Point, °C minimum	38	52	ASTM D93
2	Water and sediment volume percent, maximum	0.05	0.05	ASTM D2709
3	Physical Distillation, T90 °C, maximum	343	343	ASTM D86
4	Kinematic Viscosity, cSt at 40°C	1.3 - 4.1	1.9 - 4.1	ASTM D445

Table 7, Summary of ASTM D7467 Requirements for B6 to B20 Biodiesel Blends.				
Item	**Performance Characteristics**	**Requirements**		**Test Procedure**
- - -	- - -	**D1 Blends**	**D2 Blends**	- - -
5	Ash, mass percent, maximum	0.01	0.01	ASTM D482
6	Sulfur, wt percent, maximum	Per regulation (reference Table 1)	Per regulation (reference Table 1)	ASTM D5453, D2622, or D129, depending on sulfur content
7	Copper strip corrosion rating, maximum	Number 3	Number 3	ASTM D130
8	Cetane Number, minimum[1]	40	40	ASTM D613
9	Cloud Point[2]	Per footnote	Per footnote	ASTM D2500, D4539, D6371
10	Ramsbottom carbon residue on 10 percent distillation residue, wt percent, maximum	0.15	0.35	ASTM D524
11	Lubricity, HFRR at 60°C, micron, maximum	520	520	ASTM D6079
12	Acid number, mgKOH/g, maximum	0.3	0.3	ASTM D664
13	Biodiesel content percent (V/V)	6-20	6-20	D7371
14	Oxidation stability, induction time, hours, minimum	6	6	EN14112 (Rancimat)
15	One of the following **must** be met:			
(a)	Cetane index, minimum	40	40	D976-80

Table 7, Summary of ASTM D7467 Requirements for B6 to B20 Biodiesel Blends.				
Item	**Performance Characteristics**	**Requirements**		**Test Procedure**
- - -	**- - -**	**D1 Blends**	**D2 Blends**	**- - -**
(b)	Aromaticity, percent vol, maximum	35	35	D1319-03
[1]Low ambient temperatures, as well as operation at high altitudes may require the use of fuels with higher cetane ratings.				
[2]The maximum cloud point temperature shall be equal to or lower than the tenth percentile minimum ambient temperature in the geographical area and seasonal time frame as defined by ASTM D975.				

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